Database Management - Notes

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Session 1: Data Management

What is a Database

A database is an organized and structured set of records stored and managed for a common purpose. A data record is an information element which, as a unit, describes a complex set of facts.

Architecture and Components of an Information System

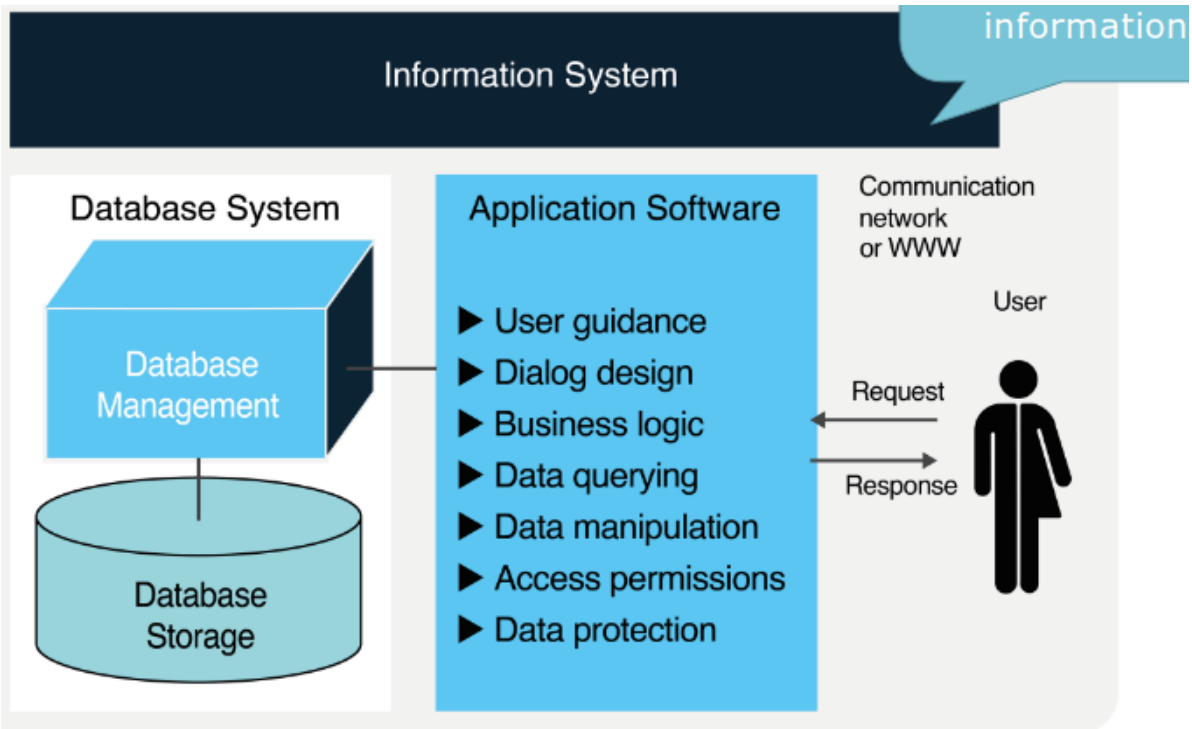

Figure 1: Information System

File System

A file system is a structured method used by an operating system to organize, store, and retrieve data on a storage drive. It manages how data is named, placed, and grouped into files and directories, acting as a bridge between the raw hardware and the user.

Why do we use databases?

- Enable multi-user operations: Manage transactions
- Query and manipulate data sets
- Ensure data consistency and integrity automatically
- Retrieve large data quantities (indexing, parallelization) efficiently

Big Data Challenges

Big data primarily refers to data sets that are too large or complex to be dealt with by traditional data-processing software. Data with many entries (rows) offers greater statistical power, while data with higher complexity (more attributes or columns) may lead to a higher false discovery rate.

- **Volume**: This refers to the sheer amount of data generated every second, requiring massive storage capacities that go far beyond traditional hardware.
- **Velocity**: This represents the incredible speed at which data is created, processed, and analyzed, often in real-time or near real-time.
- **Variety**: This describes the diverse types of data available, ranging from structured text in databases to unstructured formats like video, images, and social media posts.
- **Veracity**: This focuses on the quality, accuracy, and trustworthiness of the data, as "messy" data can lead to incorrect conclusions.
- **Value**: This is the most critical V, referring to the ability to turn bulk data into meaningful insights that drive business decisions or solve problems.

NoSQL Databases

Optimization for Big Data:

- Non-relational data structures
- Non-relational database languages

Data Management

Data Management comprises all methods related to creating value from data as a resource.

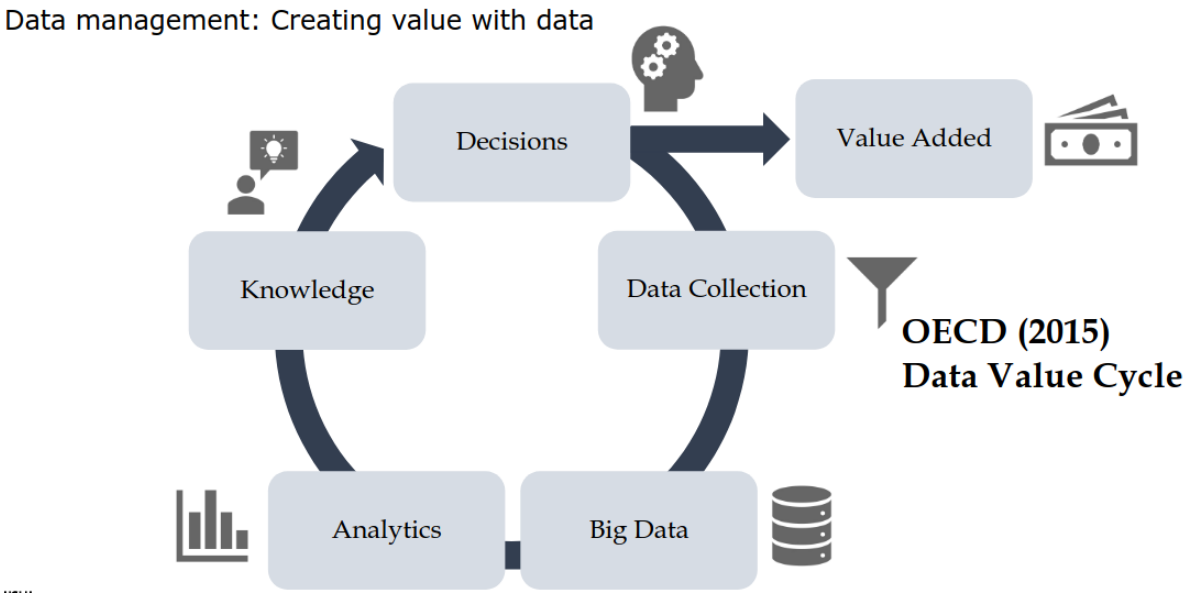

Figure 2: Data Value Cycle

Database Application

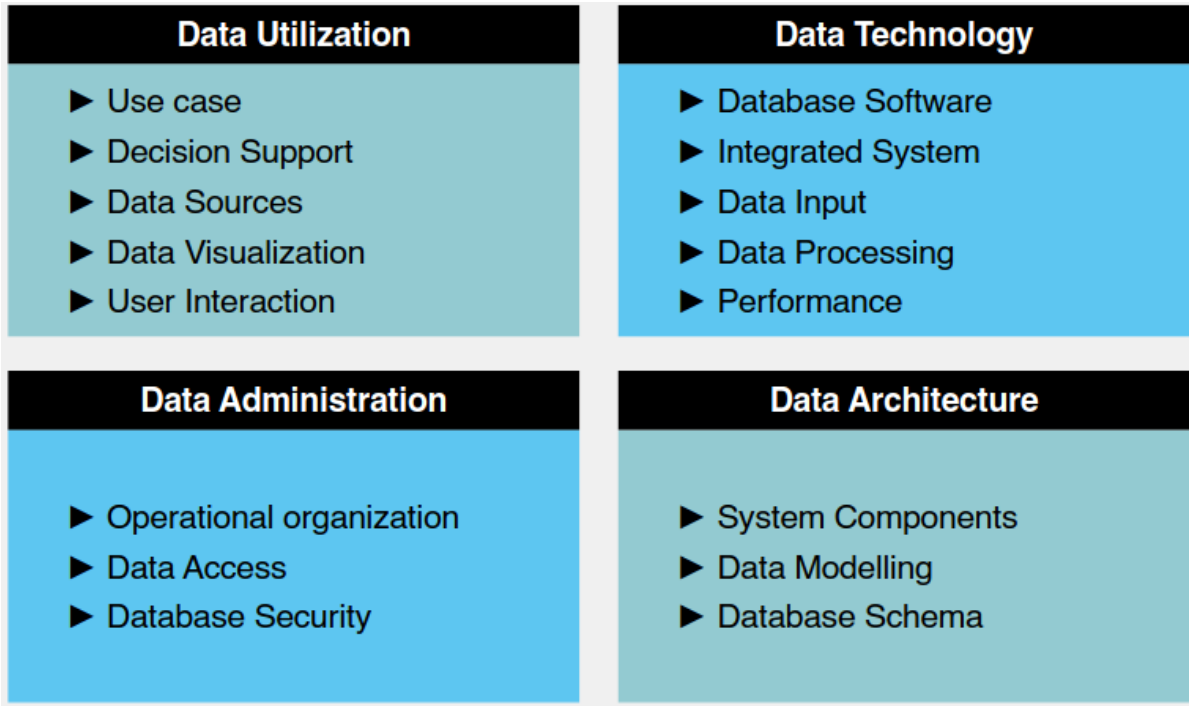

Figure 3: Database Applications

Session 2: Database Modeling

Database Modeling

Requirements Analysis

1. Requirements analysis	Order no. 113 Date: 07/14/2023
Goal	For project monitoring purposes, employees, work, and project times should periodically be reported per department.

1. Employees report to departments, with each employee being assigned to exactly one department.
2. Each project is centrally assigned a unique project number.
3. Employees can work on multiple projects simultaneously; the respective percentages of their time are logged.
4. ...

Figure 4: Requirement Analysis

Conceptual Modeling

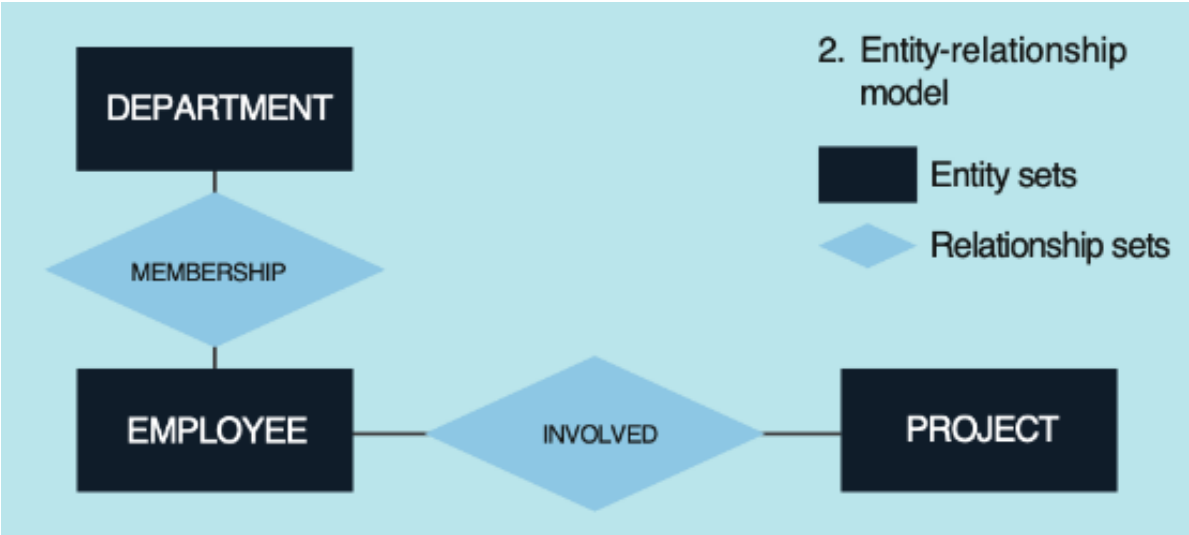

Figure 5: Conceptual Modeling

Database Shema

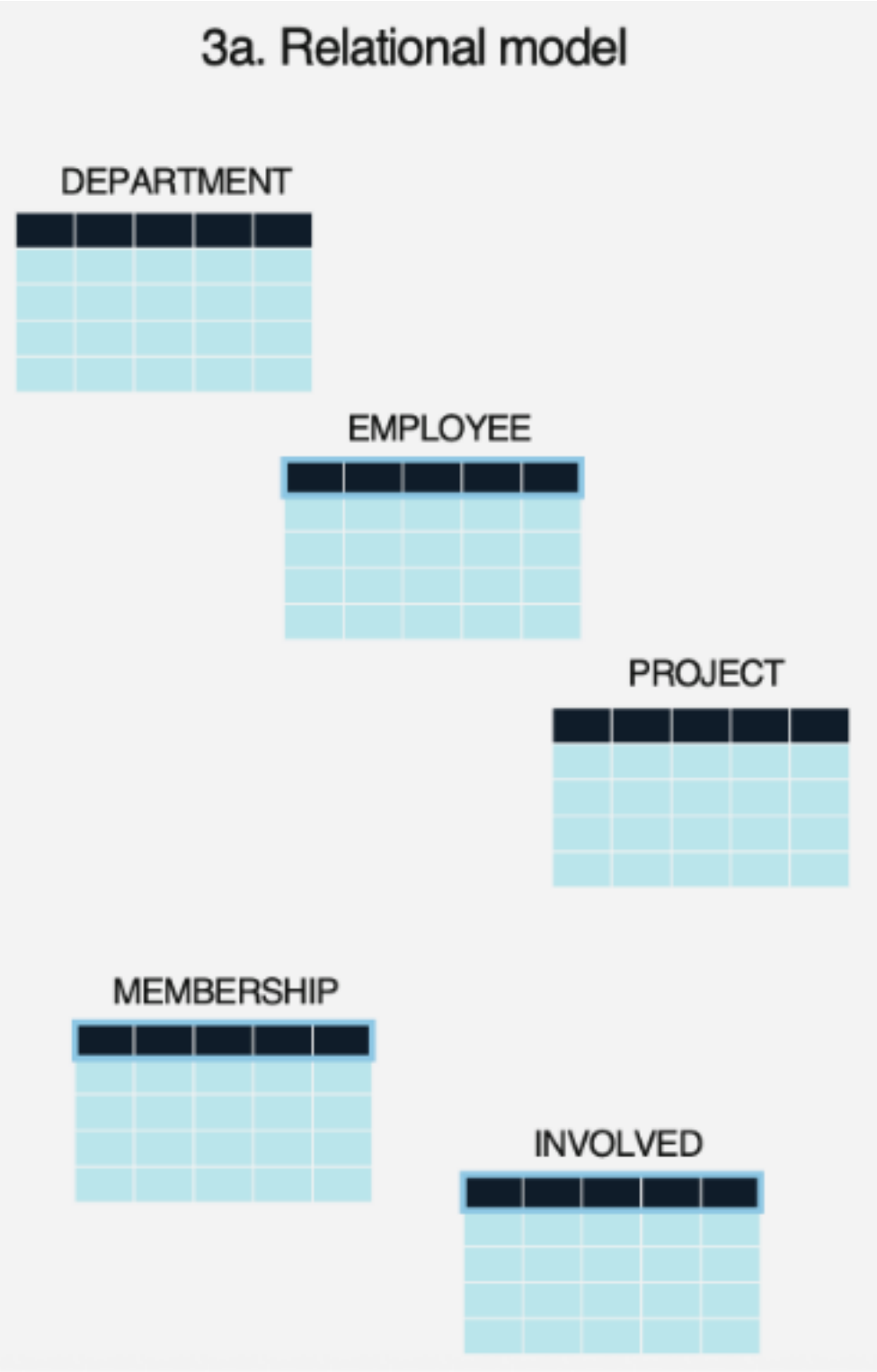

Figure 6: Database Shema

Table Defintion

A table or relation refers to a set of tuples (set of ordered, immutable objects) shown as a table.

- Table name: A table has a unambiguous name.
- Attribute name: Is unambiguous; denotes a specific column with the desired attribute.
- No column order: The attribute number is unlimited; the column order has no meaning.
- No row order: The tuple number in a table is arbitrary; the tuple order has no meaning.
- Identification key: One of the attributes or combination thereof unambiguously identifies the tuples in the table and is called the identification key

Entities

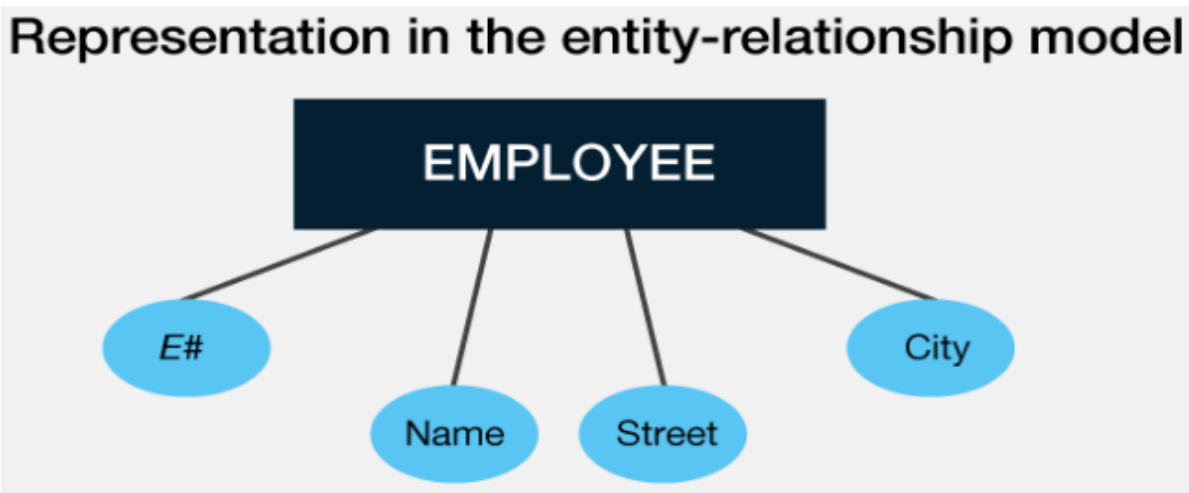

Figure 7: Table Entity

Relationships

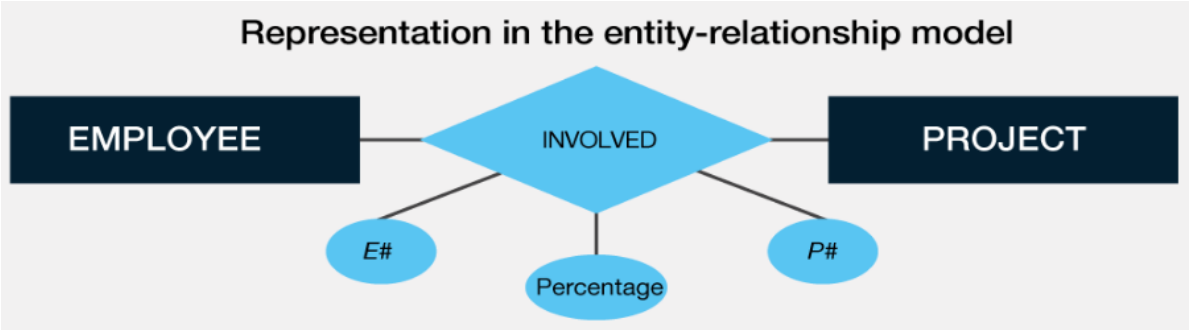

Figure 8: Entity Relationship

The Entity Relationship Model

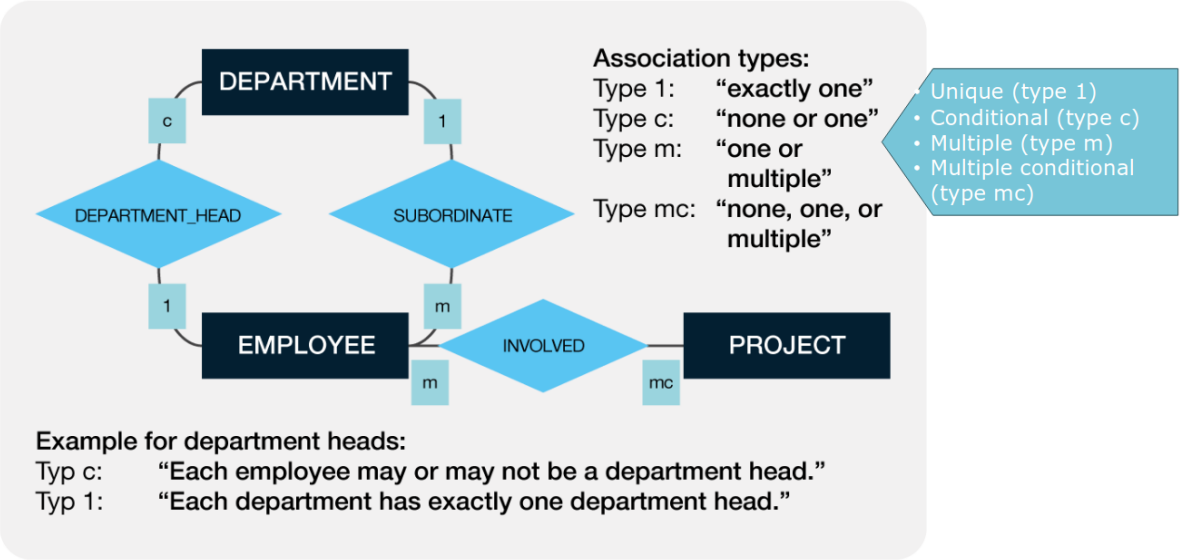

Figure 9: The Entity Relationship Model

Redundancy and Normal Forms in Tables

Redundancy in tables can lead to anomalies in databases. An attribute is redundant if individual values of it can be omitted without any loss of information. Normal forms serve to avoid redundancy.

> Note: Anomalies = Insertion anomaly, mutation anomaly, deletion anomaly.

Normal Forms

Stage	Key Focus	Goal
1NF	Atomicity	Eliminate duplicate columns and multi-valued cells.
2NF	Full Dependency	Eliminate partial dependencies on the primary key.
3NF	Direct Dependency	Eliminate dependencies between non-key columns.
4NF	Multi-valued facts	Isolate independent multi-valued relationships.
5NF	Reconstruction	Minimize redundancy in complex many-to-many relationships.

Session 3: Database Integration

SQL Data Manipulation Lanugage

- `CREATE TABLE`: Define tables
- `DROP TABLE`: Delete tables
- `ALTER TABLE`: Change table definition
- `INSERT`: Insert data

```
INSERT INTO Employee VALUES
('E22', 'Cole', '7th Street', 'Manhattan', 'A6'),
('E23', 'Smith', 'Reseda Blvd', 'Los Angeles', 'A5')
```

- `UPDATE`: Changing the data
- `SELECT`: Filter data
- `DELETE`: Delete data

Data Base Integration

Database Integration not only helps automate Data Preparation. Metadata and schema integration also contributes to data understanding.

ETL (Extract, Transform, Load)

In the traditional ETL process, data is transformed on a secondary staging server before it ever reaches the target database (usually a Data Warehouse).

> Note: Best for structured data, small-to-medium data volumes, and legacy on-premise systems.

```sql
LOAD DATA LOCAL INFILE '/path/to/ratings.csv'
INTO TABLE Ratings
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
IGNORE 1 LINES
(UserID, MovieID, Rating, @Timestamp)
SET Timestamp = FROM_UNIXTIME(@Timestamp);
```

Loading large Datasets

By Table Import Utility in the MySQL Shell.

ELT (Extract, Load, Transform)

ELT is a modern evolution where data is loaded into the target system (usually a Cloud Data Warehouse or Data Lake) in its raw form. The transformation happens within the target system using its own processing power.

> Note: Best for unstructured/Big Data, high velocity, and modern cloud environments (like Snowflake, BigQuery, or Redshift).

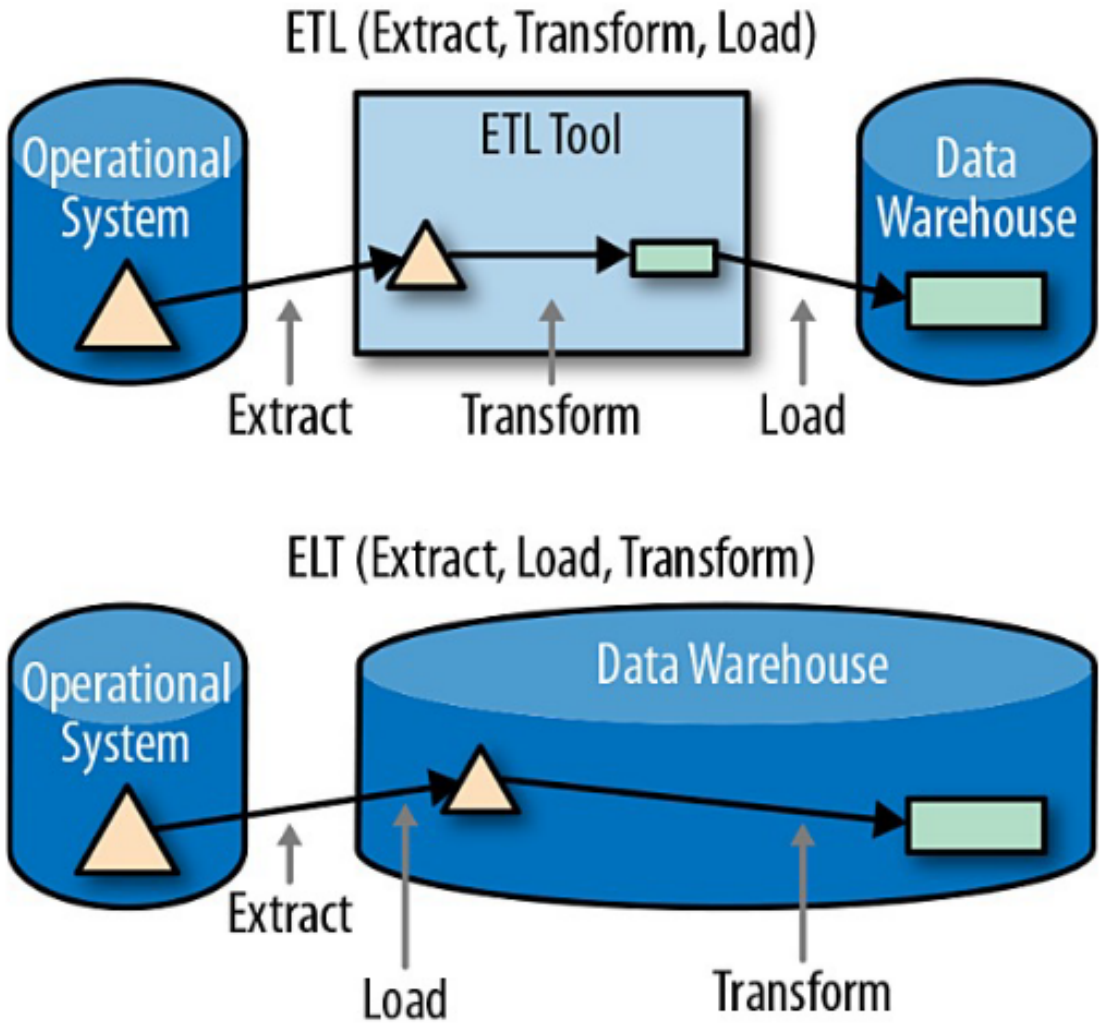

Figure 10: ETL vs. ELT